

CHAPTER 3

Precision Farming



0916CH03

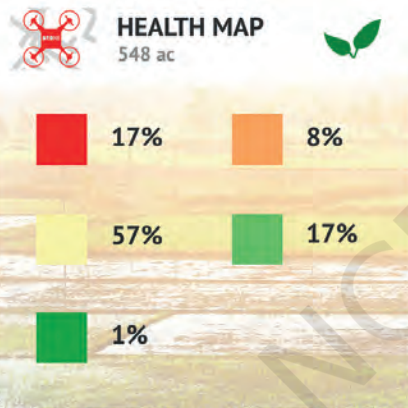
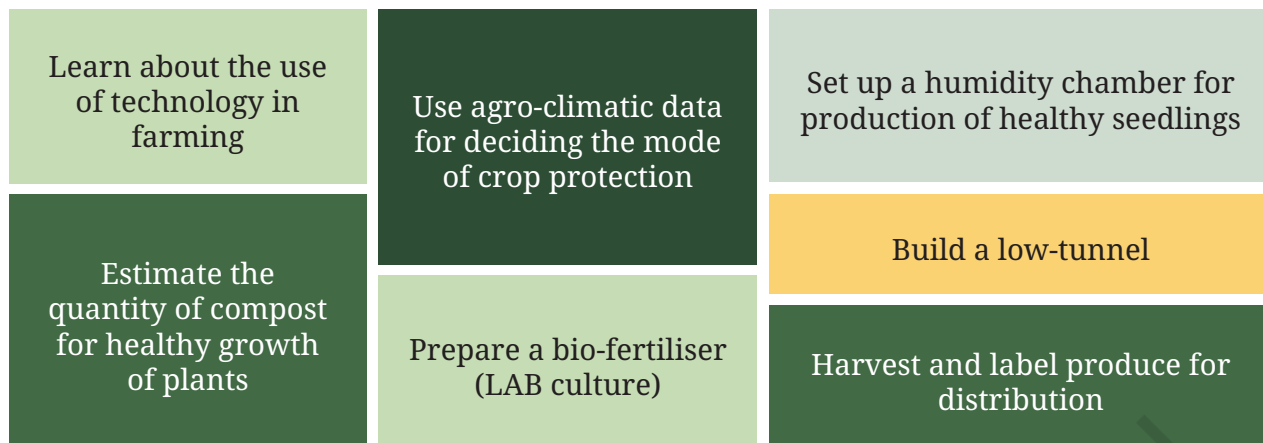


Figure 3.1: 'Health map' used for monitoring crop conditions in a field using remote sensing (satellite or drone imagery)

Different colours on the map in Figure 3.1 indicate crops health, with red indicating the worst and green indicating the best. In the figure, 17 per cent of crops are in poor health and only 1 per cent are in good health. Farmers use these data to make decisions related to irrigation, fertilisation and pest management for small areas of land.

In this chapter, you will



3.1 Introduction

Precision farming uses science and technology to improve agricultural yield, while caring for the environment. The word ‘precision’ means ‘exactly as required and consistent, that is, the same every time’. In keeping with this meaning, precision farming focuses on giving plants exactly what they need, i.e., the right amount of water, nutrients, and care at the right time and in the right amount (Table 3.1).

Precision farming is especially useful in small nurseries, since space and resources are limited. By using techniques like drip irrigation and sensors, and making data-based decisions, you can save water and avoid wastage, reduce excess use of fertilisers and pesticides, and grow healthier plants.

Greenhouses and shade-net (please refer to Grade 7 *Kaushal Bodh*) are examples of precision farming, since plants are provided with essential conditions for growth as per their need.

Table 3.1: Comparison of traditional and precision farming

Farming practice	Traditional farming	Precision farming
Agro-climatic impact	No control on climatic parameters	Precise control on temperature, humidity, light intensity, etc., through use of greenhouse, shade-net and also forecast using meteorological data.
Seed sowing	Seed broadcasting or sowing with manual tools	Sowing seeds with the help of machinery at proper spacing and depth; plant nursery management using modern techniques.

Irrigation	Flood irrigation (watering entire field)	Targeted watering using micro-irrigation (drip, sprinkler, etc.) guided by soil-moisture sensor and automated systems.
Fertiliser and pest management	Estimation of doses of fertilisers and pesticides by experience	Need-based application based on soil analysis and information from drones, satellites and apps, and experts.
Harvesting and packaging	Higher losses due to poor handling and packaging	Lower losses due to automated harvesters, using sensors and advanced packaging using digital labels.

3.2 Process chart

3.2.1 Scoping work

In order to proceed with the work, you need to make decisions about the following:

1. You can choose to work with farmers, who are already using precision farming techniques. You can request them to allow you to assist them in small groups of 5–10.
2. You can establish a precision farming unit in your school, including a small plant nursery.
3. You can convert your school garden into a precision farming unit by using technology-based inputs for decision making.

The following questions will help refine your decision-making process further:

1. Which plants should be grown? Refer to the discussion on agro-climate relationship in Chapter 1 to choose what you can grow. Ensure that the life cycle of the selected plants is 2–3 months.
2. What is useful? You need to decide what will be useful for the school or the community (for example, vegetables for the midday meal or flowers during the festival and wedding seasons).
3. Where to set up the precision farming unit? You will need a level area with good ventilation and scope for building structural support like a greenhouse. You will also need access to a reliable water source for irrigation and a separate area for storage. The entire set-up should be located, so that there is ventilation and penetration of sunlight.



Defining scope of work



- Which technology to use? For example, use of a humidity chamber, low-tunnel, shade-net or greenhouse structure, drip irrigation system, and sensors (for example, moisture sensors and climate sensors).



PORTFOLIO

- Where will you use the precision techniques – in a farmer's field or in school?
- Which plants will you grow? Justify your choice.



Making a
process chart

3.2.2 Making a process chart

A process chart lists all the tasks you will do, along with estimated dates of completion and responsibility (Table 3.2).

Table 3.2: Template for process chart

Tasks for precision farming	Dates for task	Responsibility
Site selection and layout		
Deciding crop protection method based on agro-climatic data		
Testing for organic carbon and adding required compost		
Setting up humidity chamber for raising seedlings		
Building a low-tunnel or shade-net		
Installation of drip irrigation system		
Preparing biofertiliser		
Identifying and managing pests using apps		
Harvesting		
Packaging and transport/labelling with QR code		



3.3 Site visit

To understand how precision farming works in practice, you can interact with farmers, agricultural scientists or even urban gardeners. Possible sites for visit can be Krishi Vigyan Kendra (KVK), agricultural universities or greenhouses in a community, in the presence of a teacher.



PORTFOLIO

Table 3.3 lists some points of observation, while visiting the site.

Table 3.3: Guidelines for observation during site visit

Points of observation/ discussion	Description
Tools and materials used	Materials used and their storage Tools used and their storage
Key processes	Key steps and their importance
Safety protocols	Using appropriate tools, safety precautions, etc.
Schedules	Frequency and timing of key tasks, if any
Quality criteria	Criteria for quality inputs, process and output
Technology use	Types of digital tools/apps used

You can ask the expert, what they value most about their work (for example, quality of produce, connect with nature, opportunity for employment, earnings, etc.), and how they face any specific challenges.

Create a process chart for the work you will do.



CASELET

A group of students from Grade 9 in Government High School visited farmers in their vicinity to learn about precision farming techniques such as low-tunnel farming (Figure 3.2) and drip irrigation. Using these modes farmers cultivate vegetables that usually grow in the summer and rainy season (like tomatoes and cucumbers) in harsh winter months. They learnt the process of low-tunnel farming, which involves creating tunnel-like structures made



up with hoops and polythene covers. These low-tunnels create a ‘greenhouse effect’ to protect the crops from cold weather and provide conditions necessary for their growth. Farmers also use humidity chambers, which are designed to maintain high humidity. This supports the growth of young plants that are yet to develop a root system.

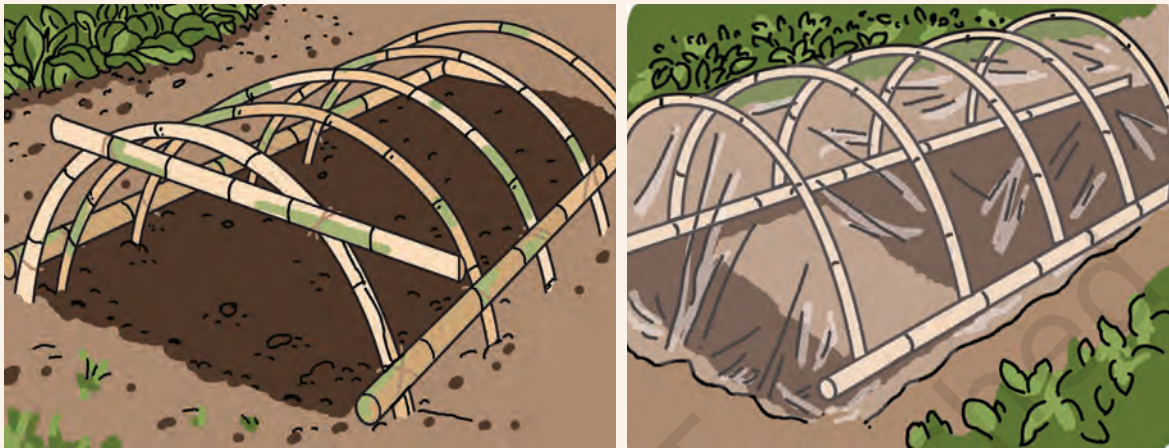


Figure 3.2: Low-tunnels used in precision farming to create necessary conditions for growth



Essential
conditions for
growth

3.4 Setting up a precision farming unit in school

3.4.1 Collecting weather data

In precision farming, you can modify climatic conditions for better plant growth by providing shade, increasing humidity, and protecting them from rain and wind. Use Table 3.4 to decide the crop protection method you will use.



TASK

Collecting weather data and deciding which crop protection method to use.

Collect data from the school meteorological observatory (Chapter 1) and identify the crop protection method for your area (Table 3.4). You can also collect data from the Indian Meteorological Department (IMD) website, local Krishi Vigyan Kendra (KVK) or an agricultural university in your locality.

Table 3.4: Deciding crop protection method

Mode of protection	Advantages
Greenhouse	Lowering temperature, protection from rainfall and frost
Low-tunnel	Increasing temperature, protection from intense sunlight
Humidity chamber	Increasing humidity, especially for nursery seedlings
Shade-net	Protection from high temperatures and scorching heat

CHECK YOUR UNDERSTANDING

Students of Government High School collected data from the school meteorological observatory and the meteorological department website. They observed that the temperature usually ranges between 22°C and 34°C (the hottest months are April and May, while December is the coolest with 22°C). The area experiences heavy monsoon rainfall, mainly from June to September (average annual precipitation of around 3,000 to 3,350 mm). Humidity remains high, between 70 and 90 per cent, especially during the monsoon season.

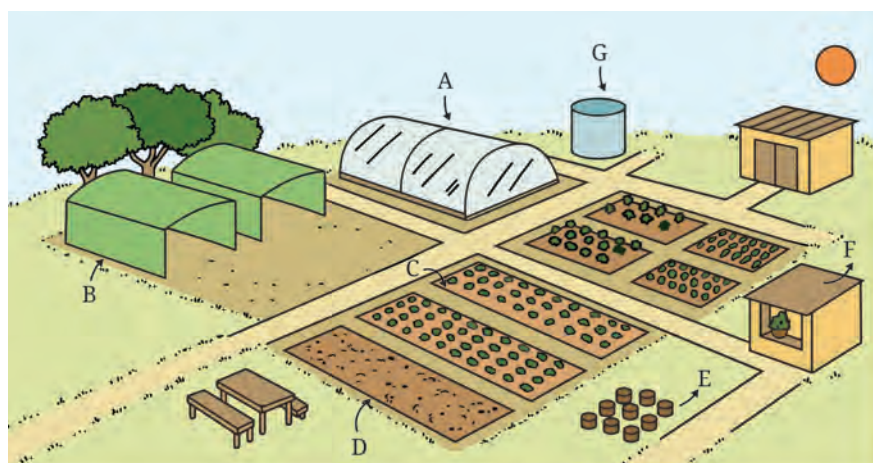
Which crop protection method should they use? Justify your selection.

3.4.2 Layout

As you design the layout of the precision farming unit, you can refer to the project on Plant Nursery in Grade 7 *Kaushal Bodh*. Figure 3.3 shows a layout that includes raised beds, low-tunnel and an area for pots.

1. On the basis of weather data and the selected plant protection mode, you can build a small greenhouse, low-tunnel or a shade-net.
2. You will also need a small space to create a humidity chamber—this provides special protection to seedlings.
3. You will need to mark areas for plant beds or to place pots for plant cultivation.
4. You will also need to indicate the placement of water tank, water pump, storage of raw material, pathways, etc.
5. You may like to keep space for composting and any other aspect you feel is important.





- A - Low tunnel
- B - Shade net (plant nursery)
- C - Raised beds
- D - Composting bed
- E - Area for pots
- F - Storage
- G - Water tank

Figure 3.3: Layout of a small precision farming unit



PORTFOLIO

Make a sketch of the layout of the precision farming unit you plan to set up. Mark the dimensions accurately.



Select tools and materials

3.5 Selecting materials

After the layout has been planned, the next step is to select materials. Table 3.5 will help you make decisions for gathering materials for the rooftop garden. You may recall some of these from Grades 6–8 *Kaushal Bodh* textbooks.

Table 3.5: Selection of materials

Materials	Use	Safety note
Seeds/saplings	For plant cultivation	Ensure viability (healthy, disease-free); handle gently to avoid crushing; store in a dry and cool place
Growing media (soil, cocopeat, vermicompost, sand and pot mix)	For seed germination and plant growth	Wear gloves, while handling to prevent skin irritation; wash hands thoroughly after handling organic media
Bamboo poles/metal frame	Framework for low-tunnel/humidity chamber/shade-net	Ensure bamboo or metal frames are securely fixed and free of sharp edges

Polyethylene sheet/ shade-net	Covering for tunnel/ humidity chamber/ shade-net	
Trays, pots and containers	Sowing of seeds and saplings	
Water hose pipe, drippers, micro-sprinkler, etc.	Irrigation	Ensure minimal wastage of water

3.6 Selecting tools

Table 3.6 below provides a list of tools necessary for developing the rooftop garden, along with their use and safe handling. You may recall some of these from the Grades 6–8 *Kaushal Bodh* textbook.

Table 3.6: Selection of tools

Tools	Use	Safety note
Hand trowel	Digging soil, planting seeds, transplanting seedlings	Do not leave sharp tools lying around; store safely after use.
Measuring tape	Marking systematic grid layout for beds and spacing	Use flexible tape to avoid sharp metal edges; store properly to prevent injury.
Hygrometer (humidity meter)	Measuring humidity in chamber	Handle carefully, as cover can break; do not expose to excess water.
Pruning scissors	Cutting stems, removing dried leaves, shaping plants	Handle carefully; keep blades closed when not in use.
Watering can/Spray bottle	Watering plants gently, especially seedlings	Avoid spilling water on the roof floor; keep a firm grip to prevent slipping.
Gloves	Protecting hands from soil, compost, and cuts	Always wear gloves, while handling soil or organic manure.
Measuring scale	Measure precise amounts	
Pipes	For drip irrigation system	
Hand tools	For making structures	As per protocols for using tools





Cost
estimation and
documentation

3.7 Making Bill of Materials

The Bill of Materials helps in estimating costs in advance, avoiding waste by buying only what is necessary and organising the work step by step. You can use the template in Table 3.7 or modify it to list your estimated materials and tools.



CASELET

Students of Government High School calculated the Bill of Materials for their work on 30 square metres of land. They planned to set up four plant beds, two for low-tunnel chambers (Figure 3.4) and two for open farming. They also planned to set up a humidity chamber, and leave space for a water tank and material storage.

With the help of the expert, they estimated the cost of tools, materials, technologies used, and the duration of the hands-on activity.



Figure 3.4: Setting up a low-tunnel for precision farming

Table 3.7: Bill of Materials

Items	Quantity	Estimated cost (in ₹)	Remarks (if any)
Trays	10	100	As per the space available on the roof
Bamboo sticks for low-tunnel frame	20 poles (about 2 m long)	200	Alternatives – wooden sticks, PVC pipes, iron rod
Mulching material	20 sq m	500	Alternatives – newspaper, used paper
Spinach seeds	250 g	150	Other alternative – seeds of any leafy vegetable
Total		950	

Cost of labour	Value (time spent in hours × hourly estimate × frequency per week)	Estimated cost (in ₹)	Remarks (if any)
Watering beds	$0.25 \times ₹10 \times 12$	30	
Marking of land and cleaning	$0.5 \text{ hr} \times ₹25 \text{ (1 time activity)}$	12.50	
Making raised beds	$2.0 \times ₹25 \times 2$	100	
Making low-tunnel	$1 \text{ hr} \times ₹25 \text{ (1 time activity)}$	25	
Total		167.50	



PORTFOLIO

Create a Bill of Materials for the work you will do.

3.8 Building a precision farming unit in school

Besides protection, a humidity chamber or low-tunnel provides a warm and humid environment, and soil moisture, thereby supporting germination and reducing need for daily watering. You can use a humidity chamber for growing seedlings and then transfer them to a low-tunnel for controlled growth of healthier plants.

3.8.1 Humidity chamber for plant nursery

A humidity chamber is a small, specialised structure that provides extremely high, precise levels of moisture and temperature, making it ideal for the rapid growth of cuttings and seed germination.



Modes of crop protection



TASK

Figure 3.5 shows three different ways to make a humidity chamber to grow saplings in a controlled environment.



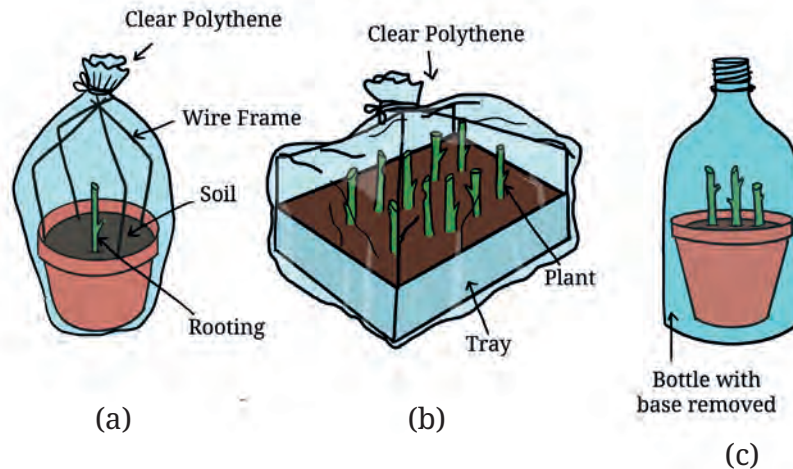


Figure 3.5: Making a humidity chamber—(a) place a clear polythene bag over a pot, (b) over a tray or (c) place a pot inside a clear bottle whose bottom has been cut off

3.8.2 Low-tunnel

Compared to a humidity chamber, a low-tunnel is a larger, less controlled structure. It primarily serves to extend the growing season by trapping solar heat and providing a protective barrier against harsh weather. This results in better crop quality and higher yields. Figure 3.6 shows how to make a simple low-tunnel.

Steps to Create a Simple Low-Tunnel



Step 1: Build a tunnel structure (frame) using bamboo, wood, or metal rods.



Step 2: Cover the frame with transparent polyethylene sheets to trap heat and moisture.



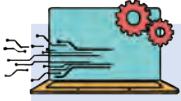
Step 3: At the base, spread a layer of sand and compost (1:1 proportion), which can be sprinkled with water to increase humidity.



Step 4: Provide (a) small openings or (b) roll-up sides for airflow to prevent overheating, as required.

Figure 3.6: Steps to create a low-tunnel

You can place trays of water or use misting bottles inside the tunnel to maintain moisture levels. You can also place a DIY dry bulb-wet bulb thermometer or use a basic hygrometer (humidity meter) to regularly check and regulate humidity.



TECHNOLOGY AND ARTIFICIAL INTELLIGENCE

Do-It-Yourself (DIY) weather monitor

To record temperature and humidity inside a humidity chamber, you can build a 'DIY weather monitor' with simple sensors and programming boards (Figure 3.7). Please refer to Project on Home Automation in Grade 8 *Kaushal Bodh*. You can also look for videos on assembling the circuit online.

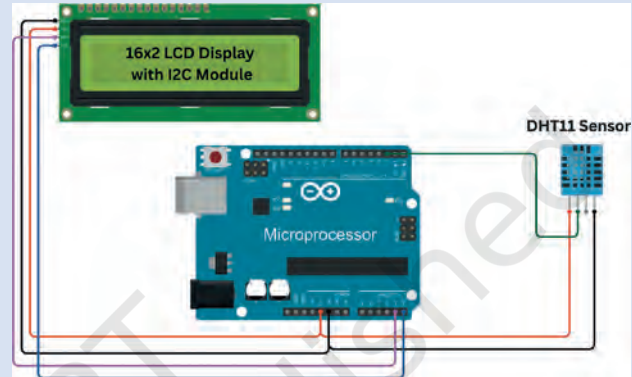


Figure 3.7: DIY weather monitor



TASK

You may have prepared plant seedlings by using plug or seedling trays and the stem cutting method (*Kaushal Bodh*, Grade 7). Now, observe the germination and growth of seedlings in a humidity chamber. Follow the simple steps for this activity:

1. Observe plants on a daily basis for germination (sprouting), growth, number of leaves, etc., and prepare an observation record.
2. Keep a record of the temperature and humidity in the humidity chamber.
3. Note any analysis or reflections, based on your observations.

3.8.3 Installation of drip irrigation system

Drip irrigation systems are used to deliver water drop by drop, directly to the root zone through drippers. Soil moisture sensors can be added to the system to ensure that water is given as per the need and when required.



Maintenance
and monitoring





TASK

Setting up drip irrigation system

Figure 3.8 provides details related to setting up a drip irrigation system.

Step 1: Laying out pipes – Arrange main pipes and smaller lateral pipes, along the plant beds.

Step 2: Attach drippers – Fix small nozzles or emitters at intervals, close to the roots of each plant.

Step 3: Connect to water source – Attach the system to a tank or pump that supplies water.

Step 4: Add filter unit – Place a simple filter to prevent clogging of pipes and drippers.

Step 5: Control flow – Use valves to adjust how much water each part of the field/garden/nursery receives.

Step 6: Integrate ‘fertiligation’ – Dissolve nutrients (fertilisers) in the irrigation tank, so that water and nutrients are delivered together in a measured amount.

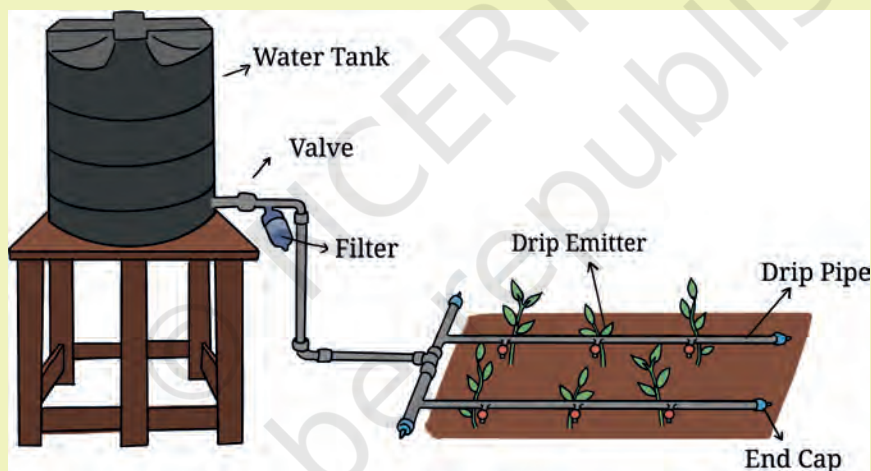
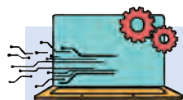


Figure 3.8: Drip irrigation system



TECHNOLOGY AND ARTIFICIAL INTELLIGENCE

Do-It-Yourself (DIY) drip irrigation system with soil-moisture sensor

In precision farming, water is provided to plants through drip irrigation (micro-irrigation). The drip system can be automated with the help of soil-moisture sensors for precise water management (Figure 3.9).

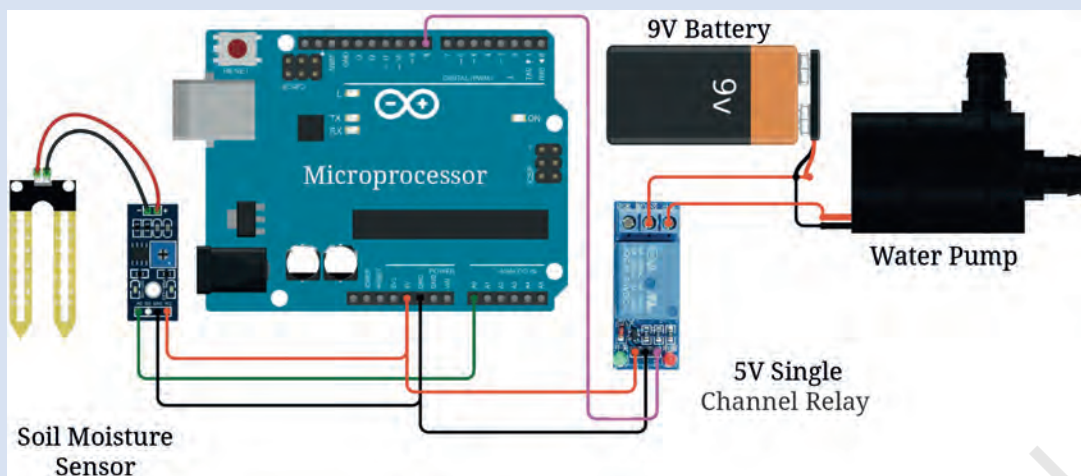


Figure 3.9: Soil–moisture sensor

3.8.4 Deciding quantity of organic fertiliser

You have learnt about testing soil for its texture (mason-jar test), and acidity or alkalinity of soil (soil pH test) in Chapter 1. In precision farming, soil nutrient management is achieved through precise measurement and supply of essential plant nutrients using a soil-sensor-based ‘fertigation’ (fertiliser supply through drip irrigation) system.

In this section, you will learn how fertiliser requirement can be estimated using a simple test.



Preparation of growing media



TASK

Rough estimation of organic carbon in soil

Take two soil samples in two small clear containers:

1. **Sample A (Control):** Soil without addition of compost
2. **Sample B (Test):** Soil enriched with compost. You will also need 3 per cent Hydrogen Peroxide (H_2O_2)

Step 1: Prepare the Samples – Place small and equal amounts of debris-free soil samples A and B in two separate containers.

Step 2: Add Hydrogen Peroxide – Pour hydrogen peroxide in both containers, such that both samples are saturated (wear safety gear during this step).

Step 3: Observe the Reaction – Observe both samples for immediate fizzing, bubbling or foaming as hydrogen peroxide reacts with the Organic Carbon (OC) in the soil (an oxidation reaction), releasing oxygen gas (O_2) (Table 3.8).



Table 3.8: Organic carbon content based on observation

Observation	Conclusion
No bubbling	Indicates poor or very low organic carbon
Light bubbling	Indicates moderate organic carbon
Intense effervescence and/or foam	Indicates good or high organic carbon

Calculating compost for maintaining from 1.5 per cent to 2 per cent organic carbon in soil

Soil must contain from 1.5 per cent to 2.0 per cent organic carbon for healthy plant growth. Organic carbon makes soil more porous for healthy root growth, enhances water-holding capacity of soil, increases nutrient availability and provides energy to beneficial microorganisms that support plant growth. Organic fertiliser is added to soil to maintain the desired amount of organic carbon.

To calculate the amount of compost required for maintaining from 1.5 per cent to 2.0 per cent organic carbon in soil, follow the steps in Figure 3.10.

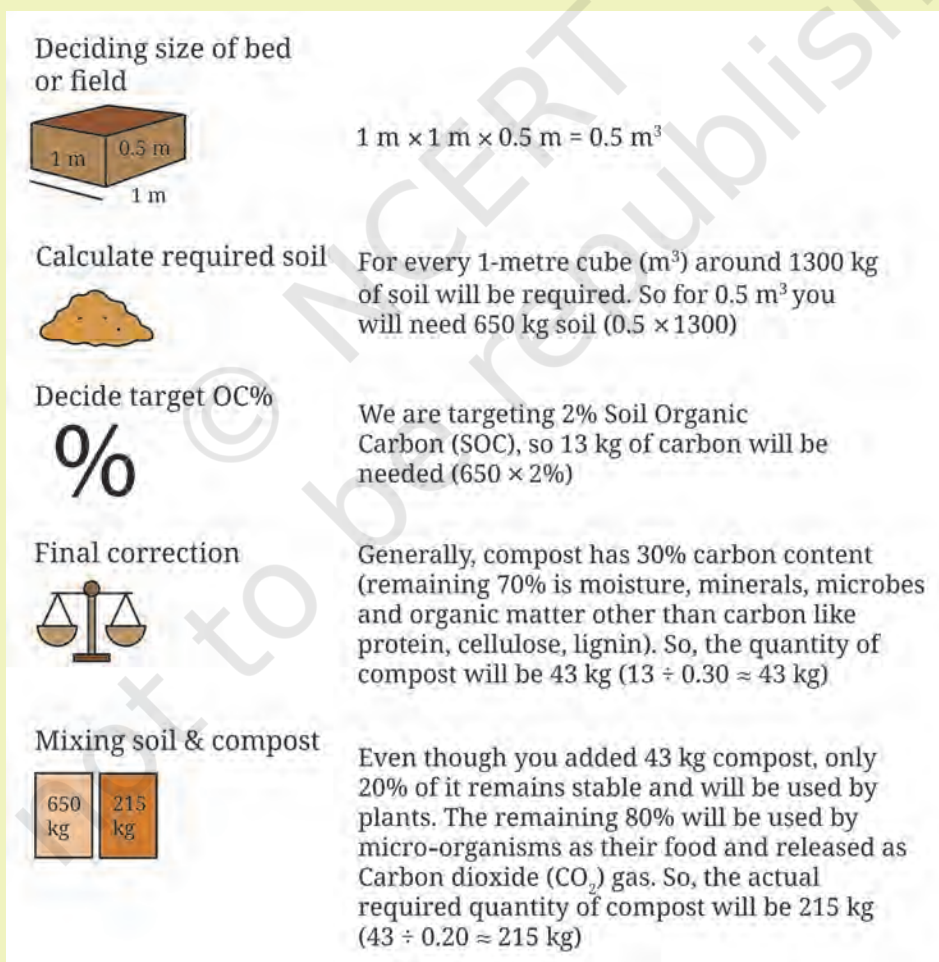


Figure 3.10: Organic carbon in soil

Once your soil and compost mix is ready with the known amount of organic carbon, you can use it for planting seedlings grown in the nursery.

3.9 Preparing an organic biofertiliser for nutrients

Plants need essential nutrients during the growth stage. These nutrients can be supplied through organic fertilisers, liquid organic manure (*jīvāmṛita*, vermiwash, *pañchagavya*, compost tea, etc.) and compost (farmyard manure, vermicompost and kitchen waste compost).

In addition to these, synthetic (chemical) fertilisers and biofertilisers are also available. Biofertilisers are made from beneficial micro-organisms, like bacterial fungi and algae. These micro-organisms help plants absorb nutrients from the soil and also improve soil fertility. You will learn how to make a biofertiliser in this section.



Maintenance and monitoring



TASK

Preparing a Lactic Acid Bacteria (LAB) culture

Lactic Acid Bacteria (LAB) culture is a useful biofertiliser containing beneficial lactic acid producing bacteria (Figure 3.11). You will need – Rice grains (500 g), clean water, milk (100 ml of any type of milk), glass jars, muslin cloth, rubber band and labels



1. Wash uncooked rice and collect the rinse water from the first two washes. Use only this water.



2. Pour the rinse water into a clean glass jar, filling it about two-thirds full. Cover the jar with a muslin cloth and secure it with a rubber band. Keep the jar at room temperature, away from direct sunlight, for 3 to 5 days; do not move it during this period.



	
3. After fermentation, a floating 'mat' will form on top. Pour out only the cloudy liquid beneath the mat. This liquid contains wild Lactic Acid Bacteria (LAB).	4. Mix one part of the fermented rice rinsed water with ten parts of milk. Pour this mixture into another clean jar, filling it, about two-thirds of the glass, and cover as before. Keep it in a dark place at room temperature for 3 to 5 days.
	
5. The jar contents will separate into solids (curds) and a yellowish liquid. Pour off the yellow liquid carefully as this is the active LAB culture. Store it in a clean bottle for use as biofertiliser.	6. Mix 1 L LAB culture in 9 L water and apply to plants through drip irrigation or direct application.

Figure 3.11: Preparing Lactic Acid Bacteria (LAB) Culture.

Remember—LAB culture has living micro-organisms (bacteria), so always use clean containers and utensils to avoid contamination. Do not shake or move the jars during fermentation, and keep the jars away from direct sunlight.



CASELET

Students of Government High School visited a nearby Krishi Vigyan Kendra (KVK) to understand different types of organic fertilisers and pesticides. They learnt that many bacteria and fungi are useful in farming. KVK's Subject Matter Expert (SME) gifted a packet of Trichoderma, a type of fungus (*Trichoderma viride*) to use as a biofertiliser. Students learnt that it is a very useful biopesticide, that lives on the surface of roots and helps the plant absorb nutrients like phosphorus. It releases lytic enzymes that dissolve the cell walls (outer skin) of harmful fungi, thereby protecting the plant. It also releases natural toxins, strong chemicals that kill harmful fungi and other pathogens.

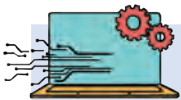
3.10 Identification and management of pests

You will need to protect your plants from harmful insect pests. As insects are part of the ecosystem, we must not eliminate them completely. Therefore, it is important to manage pest population in a way that balances crop protection with environmental health. Identification of harmful insect pests is the first step towards sustainable pest management.

You can build a simple light trap to identify various insects. A light trap is an eco-friendly pest management tool used by farmers. It uses a light source to attract and trap harmful insect pests in crop fields. Apart from identification of pests, a light trap can also help reduce the pest population.



Maintenance and monitoring



TECHNOLOGY AND ARTIFICIAL INTELLIGENCE

Do-It-Yourself (DIY)

Light Dependent Resistor (LDR) sensor for an insect trap

You can automatically switch on and switch off your insect trap during night and day, using an LDR sensor (Figure 3.12).

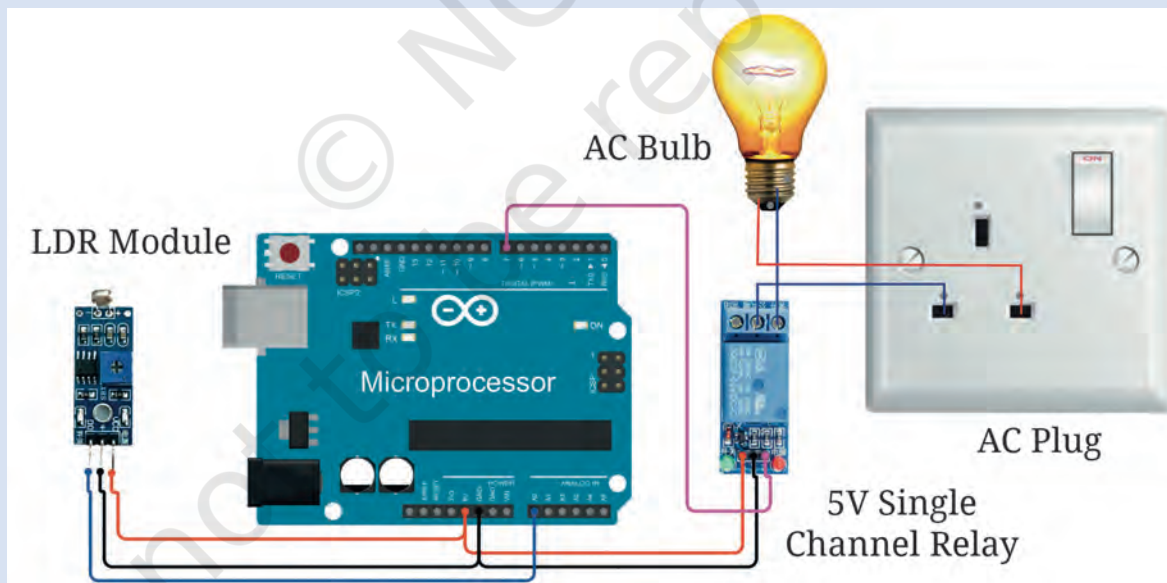


Figure 3.12: Light Dependent Resistant (LDR) sensor





TASK

Indian Council of Agricultural Research (ICAR) has launched a mobile-based pest identification and surveillance application named National Pest Surveillance System (NPSS). It uses Artificial Intelligence (AI) and Machine Learning (ML) technology for identification of insect pests in farming.

Identify the insects on the plants or those collected in the light trap using the mobile app.

1. What is their role in the ecosystem?
2. What value do they add to the ecosystem?
3. How many of them are pests (harmful for your plants)?



Harvesting,
packaging and
storage

3.11 Harvesting, packaging and storage of produce

Harvesting is the stage, where all the care and effort put into rooftop gardening yield results. The timely collection of produce is very important, because it ensures both quality and a longer shelf life of fruits, vegetables and herbs. If harvesting is delayed, the produce may become overripe, lose taste or spoil quickly. You have already learnt the basics of harvesting in Grade 6 *Kaushal Bodh* for nursery plants and Grade 8 *Kaushal Bodh* for hydroponics.

In precision farming, data-driven decisions and use of sensors help reduce crop losses, maintain quality and therefore fetch better prices.

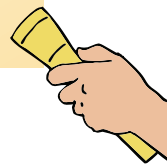
1. **Using data to determine time for harvesting:** Harvesting should be done when the temperature and humidity are the lowest. Use data (for example, temperature and humidity trends) from the DIY school meteorological observatory or the website of the meteorological department to decide the best time of day for harvesting.
2. **Climate-controlled storage:** Imagine you have just harvested a batch of fresh spinach. If the air is too dry, the leaves will wilt (become limp). If the air is too damp (high humidity) and the room is dark, mould (fungus) will start to grow. In a normal room, you would have to manually open windows or turn on fans. In precision farming, sensors do this automatically. A humidity sensor can send a signal to the drip irrigation system or a fan. Similarly,

some produce stays fresh longer in the dark since the area remains cooler – a Light Dependent Resistor can detect if a door was left open or if the sun is too bright, and automatically pull down shades or turn off the lights.

3. **Smart packaging:** Sensors to track humidity and temperature can be placed within packaging for monitoring during transport, ensuring lower losses compared to traditional methods. You can also create a QR code for the label of your package with information related to the origin of the produce, the date of harvest and the specific precision techniques (like drip irrigation) used during growth.
4. **Using data for other decisions:** Apps like Fasal and Kisan Suvidha can be used for weather updates as well as for monitoring market prices, which will help make decisions related to how long produce can be stored before selling it for the best price.

3.12 While setting up your precision farming unit

1. Use data related to agro-climatic conditions in your region for plant selection as well as making the decision – whether to use shade-net, a low-tunnel or greenhouse structure.
2. Prepare a precision nursery (humidity chamber, shade-net) for growing healthy plant seedlings.
3. Test soil and use right quantity of organic fertilisers for adjusting Soil Organic Carbon (SOC).
4. Use sensors for monitoring weather parameters (for example, temperature, humidity) and use micro-irrigation (for example, drip) for healthy plant growth.
5. Identify insect pests and diseases harming your plants and manage them by using organic pesticides.
6. Use compost and biofertiliser for healthy plants growth and yield.



3.13 Assess your learning

1. Describe the role of digital tools in precision farming. How do they change the way decisions are made on farms?
2. Create a safety checklist for the tools you used, including digital tools.
3. During a visit to a precision farming unit, list the key aspects you would observe to understand how precision techniques are applied.



4. A farmer is using random compost application in one nursery bed and measured compost in another.
5. You are asked to design a nursery layout. How would you ensure uniform growth and optimal use of resources?
6. 'With the help of technology, you can grow anything, anywhere, anytime.' Do you agree with this statement? Give two examples to support your answer. If a farmer has limited water resources, how can precision farming techniques help them use water more efficiently?
7. Suggest one low-cost innovation that could help small farmers adopt precision farming practices.
8. Of the tasks that you did, which did you enjoy the most? Which did you enjoy the least? Give examples of what went well and what did not go well. What would you do differently next time?
9. Give examples of how you can apply your learning in a real-life situation.

